

result

January 8, 2026

0.0.1 Result

[]: Final Results Table:

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Instance | Status | Primal Bound | Dual Bound | Gap (%) | Time (s) | Nodes
-----|-----|-----|-----|-----|-----|-----
30n20b8 | optimal | <adaptivediving> | +3.02000000000000e+02 | N/A | 315.54 | ┐
    ↪(Scheduler)
50v-10 | timelimit | <gins> | +3.24639796184413e+03 | N/A | 600+ | (Scheduler)
app1-2 | optimal | <rins> | -4.10000000000000e+01 | N/A | 205.98 | (Scheduler)
atlanta-ip | timelimit | <rins> | +8.25145035167828e+01 | N/A | 600+ | ┐
    ↪(Scheduler)
binkar10_1 | optimal | <rins> | +6.74220002399892e+03 | N/A | 19.53 | ┐
    ↪(Scheduler)
blp-ic98 | timelimit | <crossover> | +4.43935465363435e+03 | N/A | 600+ | ┐
    ↪(Scheduler)
eil33-2 | optimal | <relaxation> | +9.34007916000000e+02 | N/A | 77.95 | ┐
    ↪(Scheduler)
air05 | optimal | <relaxation> | +2.63740000000000e+04 | N/A | 35.79 | ┐
    ↪(Scheduler)
dano3_3 | optimal | <relaxation> | +5.76344633030644e+02 | N/A | 83.62 | ┐
    ↪(Scheduler)
mas74 | timelimit | <relaxation> | +1.14240492542426e+04 | N/A | 600+ | ┐
    ↪(Scheduler)
mas76 | optimal | <rens> | +4.00050541420000e+04 | N/A | 65.64 | (Scheduler)
mzzv11 | optimal | <relaxation> | -2.17180000000000e+04 | N/A | 255.83 | ┐
    ↪(Scheduler)
mzzv42z | optimal | <relaxation> | -2.05400000000000e+04 | N/A | 104.47 | ┐
    ↪(Scheduler)
neos5 | optimal | <objpscostdiving> | +1.50000000000000e+01 | N/A | 509.73 | ┐
    ↪(Scheduler)
neos8 | optimal | <farkasdiving> | -3.71900000000000e+03 | N/A | 5.80 | ┐
    ↪(Scheduler)
neos17 | optimal | <oneopt> | +1.50002577422577e-01 | N/A | 8.49 | (Scheduler)
nw04 | optimal | <conflictdiving> | +1.68620000000000e+04 | N/A | 33.23 | ┐
    ↪(Scheduler)
pg | optimal | <rens> | -8.67434260711703e+03 | N/A | 19.68 | (Scheduler)
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qap10 | optimal | <relaxation> | +3.39999999999997e+02 | N/A | 73.65 | ┐
    ↪(Scheduler)
roll3000 | optimal | <relaxation> | +1.28900000000000e+04 | N/A | 36.87 | ┐
    ↪(Scheduler)
seymour | timelimit | <rins> | +4.16630591749686e+02 | N/A | 600+ | (Scheduler)
timtab1 | optimal | <relaxation> | +7.64771999999880e+05 | N/A | 60.22 | ┐
    ↪(Scheduler)
uct-subprob | timelimit | <relaxation> | +2.98948096774080e+02 | N/A | 600+ | ┐
    ↪(Scheduler)
bc1 | skipped | N/A | N/A | N/A | N/A | N/A

Statistics:
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Total instances: 25
Optimal solutions: 17
Time limit reached: 6
Success rate: 68%

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0.0.2 Code

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[ ]: #!/bin/bash

echo "=== MIPLIB Benchmark with 600s Time Limit ==="
echo ""

# List of benchmark instances
INSTANCES=(
    "30n20b8"
    "50v-10"
    "app1-2"
    "atlanta-ip"
    "binkar10_1"
    "blp-ic98"
    "eil33-2"
    "air05"
    "dano3_3"
    "mas74"
    "mas76"
    "mzzv11"
    "mzzv42z"
    "neos5"
    "neos8"
    "neos17"
    "nw04"
    "pg"
    "qap10"
    "roll3000"

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    "seymour"
    "timtab1"
    "uct-subprob"
    "bc1"
)

echo "Testing ${#INSTANCES[@]} instances with 600s time limit"
echo ""

# Output files and directories
RESULTS_FILE="benchmark_600s_results.txt"
LOG_DIR="scip_logs"
mkdir -p "$LOG_DIR"

# Table header
echo "Instance | Status | Primal Bound | Dual Bound | Gap (%) | Time (s) |
↳Nodes" > "$RESULTS_FILE"
echo
↳"-----|-----|-----|-----|-----|-----|-----"
↳>> "$RESULTS_FILE"

for INSTANCE in "${INSTANCES[@]"; do
    echo -n "Running $INSTANCE (600s limit)... "

    # Check for MPS file
    MPS_FILE=""
    if [ -f "${INSTANCE}.mps" ]; then
        MPS_FILE="${INSTANCE}.mps"
    elif [ -f "${INSTANCE}.mps.gz" ]; then
        gunzip -k "${INSTANCE}.mps.gz"
        MPS_FILE="${INSTANCE}.mps"
    else
        echo "SKIP (file not found)"
        echo "$INSTANCE | skipped | N/A | N/A | N/A | N/A | N/A" >>
↳"$RESULTS_FILE"
        continue
    fi

    # Log file
    LOG_FILE="$LOG_DIR/${INSTANCE}.log"

    # Start time
    START_TIME=$(date +%s)

    # Run SCIP with a 600-second time limit (requires gtimeout from coreutils)
    gtimeout 600 scip -f "$MPS_FILE" -l "$LOG_FILE" -q

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EXIT_CODE=$?
END_TIME=$(date +%s)
ELAPSED_TIME=$((END_TIME - START_TIME))

# Parse results
if [ -f "$LOG_FILE" ]; then
    PRIMAL=$(grep "Primal Bound" "$LOG_FILE" | tail -1 | awk '{print $NF}' |
↪ || echo "N/A")
    DUAL=$(grep "Dual Bound" "$LOG_FILE" | tail -1 | awk '{print $NF}' |
↪ echo "N/A")
    NODES=$(grep "Nodes" "$LOG_FILE" | tail -1 | awk '{print $2}' |
↪ echo "N/A")
    SOLVING_TIME=$(grep "Solving Time" "$LOG_FILE" | tail -1 | awk '{print
↪ $NF}' |
↪ echo "$ELAPSED_TIME")

# Determine solver status
if [ $EXIT_CODE -eq 124 ] || [ $ELAPSED_TIME -ge 600 ]; then
    STATUS="timelimit"
    TIME_OUTPUT="600+"
elif grep -q "problem is solved" "$LOG_FILE"; then
    STATUS="optimal"
    TIME_OUTPUT="$SOLVING_TIME"
elif grep -q "problem is infeasible" "$LOG_FILE"; then
    STATUS="infeasible"
    TIME_OUTPUT="$SOLVING_TIME"
elif grep -q "unbounded" "$LOG_FILE"; then
    STATUS="unbounded"
    TIME_OUTPUT="$SOLVING_TIME"
else
    STATUS="unknown"
    TIME_OUTPUT="$ELAPSED_TIME"
fi

# Compute optimality gap
if [[ "$PRIMAL" =~ ^-[0-9]+\.[0-9]*$ ]] && [[ "$DUAL" =~ ^-[0-9]+\.[0-9]*$ ]]; then
↪ [[ "$PRIMAL" =~ ^-[0-9]+\.[0-9]*$ ]] && [[ "$DUAL" =~ ^-[0-9]+\.[0-9]*$ ]]; then
    GAP=$(echo "scale=2; (($PRIMAL - $DUAL) / $PRIMAL) * 100" | bc |
↪ echo "N/A")
else
    GAP="N/A"
fi

echo "DONE ($STATUS)"
echo "$INSTANCE | $STATUS | $PRIMAL | $DUAL | $GAP | $TIME_OUTPUT |
↪ $NODES" >> "$RESULTS_FILE"

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    if [ "$STATUS" == "optimal" ]; then
        echo "    Solved: $PRIMAL (${SOLVING_TIME}s, ${NODES} nodes)"
    elif [ "$STATUS" == "timelimit" ]; then
        echo "    Timeout: Primal=$PRIMAL, Dual=$DUAL, Gap=${GAP}%"
    fi
else
    echo "FAILED (no log)"
    echo "$INSTANCE | failed | N/A | N/A | N/A | $ELAPSED_TIME | N/A" >>_
↪"$RESULTS_FILE"
fi

# Cleanup extracted MPS file
if [ -f "${INSTANCE}.mps.gz" ] && [ -f "${INSTANCE}.mps" ]; then
    rm -f "${INSTANCE}.mps"
fi

sleep 2
done

echo ""
echo "=== Benchmark Complete ==="
echo "Results saved to: $RESULTS_FILE"
echo ""

cat "$RESULTS_FILE"

# Summary statistics
TOTAL=$(grep -c "|" "$RESULTS_FILE")
OPTIMAL=$(grep -c "optimal" "$RESULTS_FILE")
TIMELIMIT=$(grep -c "timelimit" "$RESULTS_FILE")

echo ""
echo "Statistics:"
echo "======"
echo "Total instances: $((TOTAL-1))"
echo "Optimal solutions: $OPTIMAL"
echo "Time limit reached: $TIMELIMIT"
echo "Success rate: $((OPTIMAL * 100 / (TOTAL-1)))%"

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